

1. Convex Hull
 2. Parametric Continuity Conditions
 3. Geometric Continuity Conditions
 4. Bézier Curves and Properties of Bézier Curves
 5. B-Spline Curves
 6. Beta-Splines
 7. Rational Splines
 8. Ray-Casting Methods
 9. Octrees
 10. BSP Trees
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11. Translation, Rotation, and Scaling in Three-Dimensional Transformations
 12. 3D Object Representation
 13. Categories of 3D Object Representation (List and Definitions)
 14. Perspective Projection Matrix
 15. Equations of the Perspective Projection Matrix (Explanation)
 16. General Parallel Projection Transformation
 17. Clipping Operations (Brief Explanation)
 18. Bézier Curves and Properties of Bézier Curves
 19. Steps of the Depth-Buffer Algorithm
 20. Basic Functions of the Depth-Sorting Method

Chapter 3: Output Primitives

1. Points and Lines
2. Line-Drawing Algorithms
 - DDA Algorithm
 - Bresenham's Line Algorithm
 - Parallel Line Algorithms
3. Circle-Generating Algorithms
 - Properties of Circles
 - Midpoint Circle Algorithm
4. Ellipse-Generating Algorithms
 - Properties of Ellipses
 - Midpoint Ellipse Algorithm

5. Boundary-Fill Algorithm
 6. Flood-Fill Algorithm
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Chapter 4: Attributes of Output Primitives

1. Line Attributes
 2. Curve Attributes
 3. Color and Grayscale Levels
 4. Color Tables
 5. Grayscale
 6. Character Attributes
 7. Text Attributes
 8. Pixel Phasing
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Chapter 5: Two-Dimensional Geometric Transformations

1. Basic Transformations
 - Translation
 - Rotation
 - Scaling
 2. Composite Transformations
 - Translations
 - Rotations
 - Scalings
 - General Pivot-Point Rotation
 - General Fixed-Point Scaling
 3. Other Transformations
 - Reflection
 - Shear
 4. Transformations Between Coordinate Systems
 5. Affine Transformations
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Chapter 6: Two-Dimensional Viewing

1. The Viewing Pipeline
2. Viewing Coordinate Reference Frame
3. Window-to-Viewport Coordinate Transformation
4. Two-Dimensional Viewing Functions
5. Clipping Operations
6. Point Clipping
7. Line Clipping
8. Cohen–Sutherland Line Clipping
9. Polygon Clipping
10. Sutherland–Hodgman Polygon Clipping
11. Curve Clipping
12. Text Clipping

Chapter 9

1. Three-Dimensional Display Methods
 2. Parallel Projection (*Fig. 9.4*)
 3. Perspective Projection
 4. Depth Queuing (*Fig. 9.6*)
 5. Surface Rendering
 6. Viewing Pipeline (*Fig. 9.10*)
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Chapter 10

1. Convex Hull (*Fig. 10.2*)
2. Parametric Continuity Conditions
3. Geometric Continuity Conditions
4. Bezier Curves and Its Properties
5. Spline Curves
6. Rational Splines
7. Beta-Splines
8. Ray Casting
9. Octrees

10. BSP Trees

11. Zero-Order Continuity

12. First-Order Continuity

13. Second-Order Continuity